Jayanth Vukkisa

Applied AI & Machine Learning · Data Engineering background

Hyderabad, India | +91-7093457687 | vukkisajayanth@gmail.com

vukkisa.github.io | linkedin.com/in/jayanthvukkisa | github.com/Vukkisa

SUMMARY

Postgraduate student in AI & Data Science at Jio Institute and Associate Data Scientist intern at AiSpry, building speech-recognition, retrieval-augmented generation (RAG) and computer-vision systems. Brings two and a half years as a Data Developer at Tata Consultancy Services on enterprise telecom billing data: SQL, large-scale data migration, data validation and automation.

SKILLS

Programming: Python, SQL, PL/SQL, Shell scripting
Data engineering: Oracle, PostgreSQL, data migration, ETL, data validation, CDR processing, Power BI
Machine learning and vision: PyTorch, scikit-learn, Hugging Face Transformers, YOLOv8, OpenCV, NumPy
Speech: Whisper (faster-whisper), Indic-Conformer, Qwen3-ASR, Silero VAD, WER/CER evaluation with jiwer
Generative AI: RAG, LangChain, LangGraph, Chroma, Ollama, Groq
Development tools: Django, FastAPI, Gradio, REST APIs, Git

EXPERIENCE

Associate Data Scientist Intern | AiSpry — Oct 2026 – Dec 2026

- Building a fully local speech-recognition pipeline for noisy, multilingual railway audio, within tight compute and disk limits.
- Tested denoising methods (spectral denoising, RNNoise, DeepFilterNet) and silence- and voice-based segmentation with Silero VAD.
- Benchmarked Whisper (via faster-whisper), Indic-Conformer and Qwen3-ASR side by side, and merged their outputs with a word-level ensemble.
- Scoring every variant on word and character error rate (WER, CER) with jiwer; now fine-tuning a model to lower WER. Company project: code, data and results are not public.

Data Developer | Tata Consultancy Services (TCS), Hyderabad — Sep 2023 – Apr 2026

- Configured billing rules, rate plans and system parameters for a Kenan billing platform migration of 2M+ customer records from legacy systems.
- Wrote pre-migration scripts for bulk call-detail-record (CDR) loads, improving processing speed by 25%.
- Automated a billing charge-inventory report, cutting manual validation time by 40%.
- Built compatible field mappings for financial feeds, reducing data-flow errors for downstream teams.
- Star of the Month, TCS project excellence award (2024).

PROJECTS

KnowYourRules | RAG assistant for Indian traffic law | Jio Institute team project — Sep 2026

- Built the RAG pipeline, my part of the team project: LangChain chunking, local embeddings with Ollama (nomic-embed-text) and a Chroma vector store tagged by source and state.
- Retrieves 4 central and 4 state-specific chunks per question; Groq (gpt-oss-120b) answers only from them through a strict prompt, cites its sources, or says the question isn't covered.
- Indexed the central Motor Vehicles Act and Rules, Gujarat and Delhi. Local prototype; answer accuracy not yet evaluated. github.com/DAG-21/KnowYourRules

AI Surveillance | Restricted-zone intrusion detection | Jio Institute coursework 2026

- Command-line tool: draw a polygon zone on any camera feed (webcam, USB, RTSP or video file); YOLOv8 detects people and each person's feet point is tested against the zone.
- Every intrusion saves a video clip, a snapshot and a CSV log entry. Packaged as an installable tool with a built-in FPS benchmark (results not yet recorded). github.com/Vukkisa/AI_Surveillance

Generative AI coursework | Jio Institute

- Smart Mutual Fund Advisor: RAG over a mutual-fund dataset with Chroma and structured output, served in Gradio, then rebuilt as a LangGraph graph.
- LLM extraction pipelines: LangChain and Groq pipelines that classify, summarise and extract entities from news articles and job postings.

Ez-Learn | Full-stack learning platform | Django, PostgreSQL 2023, updated 2025

- Email-OTP sign-up, course content, scored quizzes and certificates for learners; course and quiz management for instructors; Razorpay payments with verification. Deployed on Render: ez-learn.onrender.com

EDUCATION

PG Program in AI & Data Science | Jio Institute Apr 2026 – Apr 2027 (expected)

Coursework includes computer vision and generative AI.

B.Tech, Computer Science & Engineering | B.V. Raju Institute of Technology, Hyderabad 2022

CGPA 8.17/10

CERTIFICATIONS

Full Stack Developer (Python), TalentSprint, 2023: 200+ hour program in Python, Django and web development (Mar – Sep 2023)

SQL (Advanced), HackerRank, 2023